

Appendix A - Python Code

March 5, 2026

This appendix contains all Python code used for data processing, analysis, and visualization in support of the research article “Health Care Utilization and Immigration Enforcement.”

0.1 Environment Setup

Import required libraries for data processing, statistical analysis, and visualization.

```
[2]: # Import libraries
import pandas as pd
import numpy as np
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

# Display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 100)
plt.style.use('seaborn-v0_8-darkgrid')

print("Libraries imported successfully")
```

Libraries imported successfully

0.1.1 Initialize PySpark Session

Set up a Spark session for distributed data processing.

```
[ ]: # Initialize Spark session
spark = SparkSession.builder \
    .appName("Immigration_Healthcare_Analysis") \
    .config("spark.driver.memory", "4g") \
    .config("spark.executor.memory", "4g") \
    .getOrCreate()

print(f"Spark version: {spark.version}")
```

```
print(f"Spark session initialized: {spark.sparkContext.appName}")
```

0.2 2. Data Loading

Load the three primary datasets for analysis.

0.2.1 2.1 TRAC Detention Facilities Data (2019-Present)

Load average daily population data from ICE detention facilities.

```
[ ]: # Load TRAC detention facilities data
# Data source: TRAC (Transactional Records Access Clearinghouse)
# Time period: September 2019 - Present

# TODO: Update file path to actual data location
# detention_facilities = spark.read.csv('path/to/detention_facilities.csv',
    ↳header=True, inferSchema=True)

# Placeholder for demonstration
print("Load TRAC Detention Facilities data here")
# detention_facilities.printSchema()
# detention_facilities.show(5)
```

0.2.2 2.2 TRAC ICE Detention Data (2015-2019)

Load monthly counts of individuals detained by ICE at the county level with demographic breakdowns.

```
[ ]: # Load TRAC ICE detention data (county level)
# Data source: TRAC
# Time period: March 2015 - July 2019

# TODO: Update file path to actual data location
# ice_detention = spark.read.csv('path/to/ice_detention.csv', header=True,
    ↳inferSchema=True)

# Placeholder for demonstration
print("Load TRAC ICE Detention data here")
# ice_detention.printSchema()
# ice_detention.show(5)
```

0.2.3 2.3 CDC PLACES Healthcare Utilization Data

Load routine checkup data from CDC's PLACES dataset (health care outcome variable).

```
[ ]: # Load CDC PLACES data
# Data source: CDC PLACES
```

```

# Time period: 2020-2025
# Geographic levels: County, ZIP code, Census tract

# TODO: Update file path to actual data location
# places_data = spark.read.csv('path/to/places_data.csv', header=True,
    ↪inferSchema=True)

# Placeholder for demonstration
print("Load CDC PLACES healthcare utilization data here")
# places_data.printSchema()
# places_data.show(5)

```

0.3 3. Data Cleaning and Preprocessing

Clean, standardize, and prepare datasets for analysis.

```

[ ]: # Data cleaning steps:
# 1. Handle missing values
# 2. Standardize geographic identifiers (county FIPS codes, ZIP codes)
# 3. Convert date formats
# 4. Remove duplicates
# 5. Validate data quality

# Placeholder for data cleaning code
print("Data preprocessing steps will go here")

```

0.4 4. Exploratory Data Analysis

Examine basic statistics, distributions, and patterns in each dataset.

```

[ ]: # Summary statistics and initial exploration
# - Descriptive statistics for each dataset
# - Time series trends
# - Geographic distribution
# - Correlation analysis

print("Exploratory data analysis code will go here")

```

0.5 5. Panel Data Construction

Merge datasets across geography and time to create panel data for analysis.

```
[ ]: # Merge datasets on geographic identifiers (county/ZIP) and time
# Steps:
# 1. Standardize geographic keys across datasets
# 2. Align temporal granularity (monthly/yearly)
# 3. Join datasets using PySpark
# 4. Create balanced panel (if needed)

print("Panel data construction code will go here")
```

0.6 6. Statistical Modeling

Analyze the relationship between immigration enforcement intensity and healthcare utilization.

0.6.1 6.1 Model Specification

Define control variables and modeling approach.

```
[ ]: # Modeling approach:
# - Panel regression with fixed effects
# - Control variables: socioeconomic status, demographics, healthcare_
  ↪ infrastructure
# - Dependent variable: Healthcare utilization (routine checkup rates)
# - Independent variable: Immigration enforcement intensity (detention counts)
# - Subgroup analysis by citizenship status and facility type

print("Model specification code will go here")
```

0.6.2 6.2 Model Estimation

Run regression models and generate results.

```
[ ]: # Convert to pandas for statsmodels
# panel_df = panel_data.toPandas()

# Fit regression model
# model = ols('healthcare_utilization ~ enforcement_intensity + controls',
  ↪ data=panel_df).fit()

# Display results
# print(model.summary())

print("Model estimation code will go here")
```

0.7 7. Data Visualization

Create maps, scatter plots, and regression visualizations to illustrate findings.

0.7.1 7.1 Geographic Distribution Maps

Create choropleth maps showing enforcement intensity and healthcare utilization across counties.

```
[ ]: # Create geographic visualizations
# - Map 1: Immigration enforcement intensity by county
# - Map 2: Healthcare utilization rates by county
# - Map 3: Combined visualization showing correlation

print("Geographic visualization code will go here")
```

0.7.2 7.2 Scatter Plots and Regression Lines

Visualize the relationship between enforcement intensity and healthcare utilization.

```
[ ]: # Create scatter plots with regression lines
# - Overall relationship
# - By subgroups (citizenship status, facility type)
# - Time period comparisons

print("Scatter plot visualization code will go here")
```

0.7.3 7.3 Time Series Analysis

Visualize trends in enforcement and healthcare utilization over time.

```
[ ]: # Create time series visualizations
# - Trends in ICE enforcement activity (2015-2025)
# - Trends in healthcare utilization rates
# - Key policy change periods (e.g., January 2025 sensitive location policy)

print("Time series visualization code will go here")
```

0.8 8. Results Summary

Summary of key findings and implications for the research article.

```
[ ]: # Key findings:
# 1. Geographic areas with high enforcement and low healthcare utilization
# 2. Statistical significance of the relationship
# 3. Subgroup differences
# 4. Policy implications

# Export results for inclusion in research article
print("Results summary and export code will go here")

# Note: This appendix supports the findings presented in:
```

"Health Care Utilization and Immigration Enforcement" by Naomi Buell and
→ *Richie Rivera*